

EXB-05 · EXAM BLUEPRINT

Standard setting: modified Angoff and the pass mark

How a passing standard is decided, why 65 per cent is a configured mark and not yet a validated standard, and why candidates never compete with each other

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PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

— Summary

A pass mark is not a round number chosen for pass-rate optics; it is a documented judgement about what a minimally competent professional must be able to do. This document explains how the PCL-AI passing standard will be set by a modified-Angoff standard-setting study — who sits on the panel, how the minimally competent candidate is defined, what happens in each round, and how the panel's recommendation becomes a published standard. It works the arithmetic through with illustrative figures, states plainly that 65 per cent is the platform's configured mark today while the definitive standard awaits the study, and explains why criterion-referenced scoring means no candidate's result depends on anyone else's. It applies to the PCL-AI only; no examination blueprint exists yet for PFL-AI or PML-AI.

— About this document

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Standard setting: modified Angoff and the pass mark

In one paragraph. A pass mark is not a round number chosen for pass-rate optics; it is a documented judgement about what a minimally competent professional must be able to do. This document explains how the PCL-AI passing standard will be set by a modified-Angoff standard-setting study — who sits on the panel, how the minimally competent candidate is defined, what happens in each round, and how the panel's recommendation becomes a published standard. It works the arithmetic through with illustrative figures, states plainly that 65 per cent is the platform's configured mark today while the definitive standard awaits the study, and explains why criterion-referenced scoring means no candidate's result depends on anyone else's. It applies to the PCL-AI only; no examination blueprint exists yet for PFL-AI or PML-AI.

Who this is for. Candidates who want to know what the pass mark actually represents; employers deciding what a PCL-AI pass evidences; and assessment professionals auditing whether the Institute's standard is set by a method or by a preference.

— 1. Where the pass mark comes from — the honest position

Two statements are both true, and the Institute publishes both because a candidate is entitled to know which one they are being measured against.

65 per cent is the platform's configured pass mark. It is what the examination applies today, it is what appears in the candidate handbook, and it is the figure against which a result is currently decided.

The definitive standard will be set by a modified-Angoff standard-setting study. Until that study reports, 65 per cent is a configured operating value — a defensible starting point, chosen in the same region as long-standing practice in professional certification — and not a validated standard derived from this examination's own content.

Those are different things. A configured value is an administrative setting. A validated standard is a documented judgement about competence, made by qualified people, on this examination's actual items, with a record that can be audited afterwards. Presenting the first as though it were the second is the specific misrepresentation this document exists to avoid.

— 2. What a passing standard is a judgement about

A pass mark answers a question that sounds simple and is not: **how much of this examination must a candidate answer correctly to demonstrate the competence the credential attests?**

Notice what the question is not. It is not how many candidates should pass — that is a norm-referenced question, and the Institute does not ask it. It is not what mark feels rigorous — that is aes-

thetics. And it is not what did we use last time — that is inheritance, which is how a great many pass marks in professional certification were actually arrived at.

The question is about the content. Standard setting answers it by having qualified people examine the actual items and judge, item by item, what a just-competent professional would do with them.

— 3. Criterion-referenced: why candidates do not compete

The PCL-AI is **criterion-referenced, not graded on a curve**. Everyone who demonstrates the required standard passes, however many others do. There is no quota, no fixed proportion who succeed or fail, and no ranking against other candidates.

The difference is not academic. Illustrative figures.

Consider two candidates who give **identical** answers to an identical form, sitting in different months.

Under a **norm-referenced** design in which, say, the top 30 per cent of each cohort passes, the outcome depends on the cohort. In a month when the candidate sits alongside an unusually strong group, identical performance ranks below the thirtieth percentile and fails. In a weaker month, it ranks above and passes. The candidate's competence has not changed; only the company they kept has.

Under the **criterion-referenced** design the Institute uses, identical performance produces an identical outcome, in every month, in every cohort, on every form assembled to the blueprint. The comparison is against a published standard, not against other people.

Three consequences follow, and they are worth stating because candidates ask about all three.

A resit clears exactly the bar a first sitting cleared. The standard does not rise because you failed once, and it does not fall because the cohort is small.

There is no advantage in sitting at a quiet time of year, and no disadvantage in sitting alongside a strong cohort.

The Institute publishes no target pass rate. A pass rate is an outcome of a criterion-referenced system, not an input to it. A certification body that manages its pass rate towards a number has stopped running a criterion-referenced examination, whatever its documentation says.

— 4. The minimally competent candidate

Everything in standard setting rests on one construct, and getting it wrong invalidates the whole study.

The **minimally competent candidate** is the person who is just good enough. Not the excellent candidate, not the average candidate, not the candidate the panellists would want to hire — the one who sits exactly at the boundary the credential is drawing. They would be safe to certify; anyone slightly weaker would not.

Panels find this genuinely hard, and predictably so. Experienced practitioners tend to picture a good professional, because that is who they work with and who they would recommend. Picturing a good professional and rating items against them sets the standard too high — and it does so invisibly, because every rating looks reasonable on its own.

The study therefore spends its first substantial block of time not rating anything, but building a shared, written description of this person: what a just-competent PCL-AI holder can do, what they would still need to check, what they would escalate, and what they would get wrong. That description is written down, agreed, and referred back to whenever ratings diverge. A panel that has not written it down is rating against several different people at once and does not know it.

— 5. Panel composition

A standard-setting panel is a measurement instrument, and its composition determines what it can measure.

The Institute's requirements for the PCL-AI panel:

Subject-matter expertise across the whole blueprint. Panellists between them must cover project accounting and finance, project management principles, and AI in project controls. A panel with no credible AI practitioner cannot rate Domain 13 items, and will systematically misjudge them in one direction or the other.

Familiarity with the target population. Panellists must know the people who will actually sit the examination — what a practitioner with three years' experience in any field typically knows and does not know. Expertise without that familiarity produces a standard set for a population that does not exist.

Diversity of region and sector. Project controls is practised differently in construction, energy, infrastructure, manufacturing, defence and technology delivery, and differently again across regions. A single-sector panel encodes its own sector's norms as the standard.

Independence. Panellists do not rate items they wrote, and the panel is chaired by someone who does not rate at all — the chair manages process, not content. Anyone with a commercial interest in the pass rate, including training providers, is excluded from rating.

Trained on the method. Panellists are trained on the task, practise on items outside the study, and demonstrate they understand the rating question before the study begins. Untrained panels produce ratings that look like data and are not.

Panel size and precise composition: [\[CONFIRM: standard-setting panel size and composition\]](#). The study date: [\[CONFIRM: date of the standard-setting study\]](#).

— 6. The method

The Angoff method takes its name from the psychometrician William Angoff. **Modified** distinguishes the multi-round, feedback-informed variants now in general use from the original single-round formulation.

6.1 The rating question

For each item, every panellist independently answers one question:

Out of 100 minimally competent candidates, how many would answer this item correctly?

The answer is a number from 0 to 100. It is **not** a judgement of whether the item is good, whether the panellist likes it, or whether the panellist would answer it correctly. It is an estimate about a specific imagined population — the one described in §4.

Two disciplines apply. Panellists rate against the written description of the minimally competent candidate, not against themselves. And because a four-option item has a chance floor of 25 per cent ($1 \div 4$), a rating materially below 25 flags the item rather than the candidate: it says the panellist believes minimally competent candidates would do worse than random guessing, which is a statement about a defective item.

6.2 Round one — independent

Panellists rate every item alone, without discussion. Independence matters: a first round contaminated by the loudest voice in the room produces one person's standard wearing a panel's clothes.

6.3 Feedback and discussion

The panel is then shown, item by item, the spread of its own ratings — the mean and the range, without attribution. Discussion focuses on the items where the spread is widest, because a wide spread means panellists were rating different people or reading the item differently, and both are fixable.

Where empirical data exists — the item statistics from [EXB-04 – How items are written, reviewed and retired](#) §7 — it is introduced at this point. Panellists see how candidates actually performed on the item. This is one of the two things that make the method "modified", and it is a substantial improvement: it anchors judgement without replacing it, and it catches the item that everyone on the panel thought was easy and no candidate could answer.

Discussion is not a negotiation towards consensus. Nobody is asked to change a rating, and a panellist who holds their original view after hearing the discussion is doing the task correctly.

6.4 Round two — revised

Panellists rate again, independently, having heard the discussion and seen the data. The round-two ratings are the ones that count.

6.5 Computing the cut score

For each item, the item cut score is the mean of the panellists' round-two ratings. The recommended cut score for the form is the mean of the item cut scores — equivalently, the sum of the item means divided by the number of items.

Because every scored item carries equal weight (see [EXB-03 – Cognitive levels and item types](#) §4.2), that figure is directly a percentage of the form.

— 7. Worked example

Illustrative figures. Five panellists, six items, round-two ratings. Ratings are the estimated percentage of minimally competent candidates answering correctly. Means are rounded to one decimal place; the final cut score to one decimal place.

Item	P1	P2	P3	P4	P5	Sum	Item cut score (mean)	Range
1	80	75	85	70	80	390	78.0	15
2	60	55	65	50	60	290	58.0	15
3	45	40	55	35	50	225	45.0	20
4	70	65	75	70	65	345	69.0	10
5	55	60	50	60	55	280	56.0	10
6	85	90	80	85	85	425	85.0	10
Total of item means							391.0	

Recommended cut score = sum of item cut scores ÷ number of items = 391.0 ÷ 6 = 65.2 %

Read the table as a panel chair would.

Item 6 is easy for this population — the panel expects 85 per cent of minimally competent candidates to get it. That is not a criticism. An examination needs items across the difficulty range, and an item this easy still contributes if it discriminates.

Item 3 is the one to discuss. Its ratings span 35 to 55, the widest range on the form, and it sits closest to the chance floor. A range that wide after a feedback round says panellists are still reading the item differently — which usually means the item is ambiguous, not that the panellists are careless. This is the item the chair sends back to the Assessment Lead for technical review, and if it is withdrawn the cut score is recomputed on the remaining items.

The result is a recommendation, not a decision. 65.2 per cent is what this panel, on this form, recommends. What happens next is §8.

— 8. From recommendation to published standard

A standard-setting study produces a number with error attached, and pretending otherwise is the most common way a good study produces a bad decision.

Measurement error is computed and stated. Every score carries a standard error of measurement — the spread of results the same candidate would produce on repeated equivalent sittings. It is computed from the form's own statistics, so it cannot be stated in advance of the form, and any body quoting one before its examination exists is quoting a guess. What the Institute commits to is that the error will be computed, published with the study, and taken into account by a **decision rule fixed before the study runs** — not selected afterwards once the panel's number is known, which is the point at which a technical adjustment becomes a thumb on the scale.

The decision belongs to governance, not to the panel. The panel recommends; the Chief Examiner and the Certification & Impartiality Committee decide. The decision may adopt the recommendation, or depart from it for reasons that are recorded in writing. What it may not do is depart from it silently, or for reasons related to the pass rate the standard would produce.

Everything is documented. The study report records the panel composition and how it was selected, the training given, the written description of the minimally competent candidate, both rounds of ratings, the discussion record, the empirical data introduced, the computed cut score, the meas-

urement error, the decision rule and the final decision with its reasons. That report is what makes the standard defensible in an appeal, and its existence is the difference between a standard and an assertion.

Forms are held to the same standard, not to the same raw score. A standard is a level of competence. If two forms differ in difficulty, holding the raw percentage constant would silently move the standard, so the raw passing score must be adjusted to keep the standard where it is. Today the platform applies a single configured percentage to every form. Once item statistics support it, any form-level adjustment the Institute adopts will be published in this document rather than applied quietly — because a candidate who discovers after the fact that forms were not equated has been told something untrue about comparability.

— 9. If the study moves the mark

The study may put the standard somewhere other than 65 per cent. That is the point of running it, and the Institute's commitments if it happens are stated now, before there is any incentive to word them conveniently.

The change is published, with the study's method, panel composition and reasoning.

It applies from a stated effective date, with notice. Candidates who have booked against a published standard sit against that standard.

It is never applied retrospectively to results already issued. A credential earned against the standard in force at the sitting remains earned.

It is published in either direction. A standard that moves down is published exactly as a standard that moves up. A body that only announces its increases is managing perception, not standards.

— 10. How standard setting goes wrong

Rating against the good candidate. The dominant failure mode, described in §4. It inflates the standard and it is invisible without a written description of the minimally competent candidate to rate against.

Choosing the number first. A panel convened to justify 65 per cent will find its way to approximately 65 per cent. The order matters: define the candidate, rate the items, compute the result, then decide — and publish the result even when it is inconvenient.

A panel that is really one person. Without independent first-round ratings and unattributed feedback, the most senior or most confident voice sets the standard and five signatures make it look like a panel.

Introducing empirical data too early. Show panellists the item statistics before they have formed their own judgement and they anchor on the data, which turns a criterion-referenced study into a norm-referenced one wearing a disguise.

Treating the cut score as exact. 65.2 per cent is not meaningfully different from 65.4 per cent, and a process that behaves as though it were is over-reading its own precision. This is what the measurement error in §8 exists to discipline.

Never revisiting it. A standard set once and inherited for a decade is a standard set for a discipline that no longer exists — a particular risk where Domain 13 content ages fastest.

Letting the pass rate lead. The moment a pass rate becomes a target, the standard has been abandoned. This is the failure that is hardest to detect from outside and easiest to justify from inside.

— 11. Checklist

For anyone auditing a standard-setting study — ours or another body's:

- [] Is there a written description of the minimally competent candidate, agreed before rating began?
 - [] Were panellists trained on the method and given practice items outside the study?
 - [] Was round one rated independently, before any discussion?
 - [] Was feedback unattributed, and focused on items with the widest spread?
 - [] Was empirical item data introduced after round one, not before?
 - [] Were panellists free to keep their original ratings after discussion?
 - [] Was the decision rule for measurement error fixed before the panel's number was known?
 - [] Does the report name the panel's composition, and does the composition cover the whole blueprint?
 - [] Were people with a commercial interest in the pass rate excluded from rating?
 - [] Is the study documented well enough that an appeal could be answered from it years later?
 - [] Is the resulting standard published, including if it moved downwards?
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The reason to describe a standard-setting study in this much detail before running one is that the details are where standards are quietly lost. Any body can name the method; the method is not the safeguard. The safeguards are the written description of the borderline candidate, the independent first round, the decision rule fixed in advance, and the willingness to publish a number that is lower than the one you started with. When the PCL-AI study reports, this document is the specification it will be judged against.

— Related

- [EXB-01 – Examination blueprint – PCL-AI](#) — the configured pass mark and everything else the specification fixes
 - [EXB-04 – How items are written, reviewed and retired](#) — the item statistics introduced in the feedback round
 - [EXB-06 – Job-task analysis: where the blueprint gets its authority](#) — the study that defines the competence this standard is set against
 - [CER-07 – Results, scoring, appeals and complaints](#) — what a candidate does if they disagree with a result
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— Sources and standards

The modified-Angoff method is named after the psychometrician William Angoff and is described here in the Institute's own words; no published text is reproduced. The illustrative ratings in §7 are invented for teaching and were independently recomputed before submission — they are not data from any study, panel or candidate population. The certification framework is developed with reference to ISO/IEC 17024 personnel-certification principles; PCI is not accredited by any ISO/IEC 17024 accreditation body.

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